ITA0329-Mobile Computing

C01-AT2-Short Answer test

Date:13/10/2026 **Name:** T Praveen Kumar

Reg.No: 192472234

Q1. Full-Duplex Channels

Formula: Channels = Total Bandwidth / (2 × Channel Bandwidth)

Calculation: 45000 / (2 × 25) = 900 channels.

Q2. Channels Per Cell

Calculation: 560 / 7 = 80 channels per cell.

Q3. Total GSM Voice Channels

Calculation: 100 × 8 = 800 voice channels.

Q4. Power Ratio in dB

Calculation: $10 \log_{10}(80 / 4) = 13.01$ dB.

Q5. Carrier-to-Interference Ratio

Calculation: -60 - (-92) = 32 dB.

Q6. Channels Per Cell

Calculation: 42000 / 300 / 7 = 20 channels per cell.

Q7. Received Power

Calculation: $64 / 2^4 = 4$ W.

Q8. Transmit Power Increase

Calculation: $2^2 = 4$ times the original power.

Q9. Total Cluster Channels

Calculation: 4 × 15 = 60 channels.

Q11. TDMA Time Slot

Calculation: 5 / 8 = 0.625 ms.

Q12. Carrier Frequencies

Calculation: 24000 / 200 = 120 carriers.

Q14. Free-Space Path Loss

Calculation: $32.44 + 20 \log_{10}(2.5) + 20 \log_{10}(1900) = 105.98$ dB.

Q15. Call-Handling Capacity

Calculation: 6 × 250 = 1500 calls.

Q17. Total Bandwidth

Calculation: 150 × 25 = 3750 kHz = 3.75 MHz.

Q18. Guard-Band Bandwidth

Calculation: (30 - 1) × 40 = 1160 kHz = 1.16 MHz.

Q20. Total User Capacity

Calculation: 8 × 40 × 250 = 80,000 users.

Q22. FDMA Channels

Calculation: 6000 / 30 = 200 channels.

Q23. Offered Traffic

Formula: Traffic = (Subscribers × Calls per Hour × Call Duration) / 60

Calculation: (1200 × 4 × 2.5) / 60 = 200 Erlangs.

Q24. Maximum Simultaneous Users

Calculation: 40000 / 400 = 100 users.

Q25. Total Network Capacity

Calculation: 1200 × 600 = 720,000 users.

Q26. Paging Rounds

Calculation: 200 / 4 = 50 rounds.

Q28. Roaming Subscribers

Calculation: 600000 × 15 / 100 = 90,000 subscribers.

Q29. Switching Center Utilization

Formula: Utilization = (Actual Load / Total Capacity) × 100

Calculation: (9000 / 12000) × 100 = 75%.

Q32. Successful Handoffs

Calculation: 2500 × 99 / 100 = 2475 successful handoffs.

Q34. Cluster User Capacity

Calculation: 7 × 900 = 6300 users.